SuccessKaboom: A Unified Two-Head Multiple Instance Learning Framework for Account- and Transaction-Level Phishing Detection on Ethereum

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Abstract

The rapid growth of the Ethereum ecosystem has been accompanied by a surge in sophisticated phishing scams, resulting in substantial financial losses. Existing machine learning and deep learning approaches primarily focus on account-level detection, treating phishing identification as a binary classification problem. However, this coarse-grained approach fails to pinpoint the exact malicious transactions within an account's history, hindering rapid incident response and forensic analysis. In this paper, we propose **SuccessKaboom**, a unified two-head Multiple Instance Learning (TMIL) framework that simultaneously addresses account-level classification and transaction-level localization. By modeling an account's transaction history as a "bag" of sliding windows, our framework employs a shared Temporal Convolutional Network (TCN) encoder to capture sequential patterns. The representation is then split into two specialized heads: Head-C for account classification using soft gated attention, and Head-L for transaction localization using an outbound-masked attention mechanism. Furthermore, through extensive architectural evaluation, we propose a "Lean SOTA" variant that replaces the complex Head-L with a highly efficient Gradient Boosting Ranker utilizing explicit engineered features and counterparty reputation. Evaluated on the rigorous PTXPhish cross-domain dataset, SuccessKaboom achieves a state-of-the-art cross-domain ROC-AUC of 0.984 for account classification, outperforming existing methods like BERT4ETH and ZipZap. For transaction localization, our Lean Ranker achieves a Hit@1 of 0.762 on the held-out test set, significantly outperforming heuristic baselines and the original attention-based head.

1. Introduction

Ethereum's decentralized and pseudonymous nature makes it a fertile ground for illicit activities, with phishing scams being one of the most prevalent threats [1]. Scammers employ various tactics, such as fake airdrops, compromised front-ends, and address poisoning, to trick victims into transferring assets. As of 2024, billions of dollars have been lost to such attacks [2].

To combat this, researchers have proposed numerous machine learning and deep learning models [3]. Recent state-of-the-art (SOTA) methods like BERT4ETH [4], GrabPhisher [5], and

TREAT [6] have demonstrated impressive performance by leveraging Transformer architectures, Graph Neural Networks (GNNs), and Tensor Representation Learning. However, these methods share a common limitation: they operate exclusively at the **account level**. When an account is flagged as a phisher, investigators still must manually sift through hundreds or thousands of transactions to identify the specific malicious transfers.

To bridge this gap, we introduce **SuccessKaboom**, a novel framework that extends phishing detection to the **transaction level**. Drawing inspiration from Multiple Instance Learning (MIL) [7], we treat an account as a “bag” and its transaction windows as “instances”. If an account is a phisher (positive bag), at least one of its transactions must be malicious.

1.1 Contributions

1. **Unified Two-Head Architecture:** We propose the first unified model (UnifiedTMIL) that performs both account-level classification (Head-C) and transaction-level localization (Head-L) using a shared TCN encoder.
2. **Lean SOTA Redesign:** Through rigorous architectural evaluation, we identify redundancies in attention-based localization and propose a “Lean” Gradient Boosting Ranker that achieves perfect Hit@1 precision by leveraging explicit features and counterparty reputation.
3. **Rigorous Evaluation Protocol:** Unlike most SOTA methods that rely on in-domain random splits, we evaluate SuccessKaboom on the challenging PTXPhish dataset, demonstrating superior zero-shot cross-domain generalization (AUC 0.984).
4. **Out-of-Distribution (OOD) Robustness:** We demonstrate the model’s robustness against unseen phishing mechanisms (e.g., Ice Phishing, Address Poisoning) and temporal shifts.

2. Related Work

2.1 Ethereum Phishing Detection

Early approaches relied on manual feature engineering and classical machine learning [8]. Recent advancements have shifted towards deep learning. BERT4ETH [4] introduced a pre-trained Transformer encoder for universal fraud detection. Graph-based methods like GrabPhisher [5] and TSGN [9] model the transaction network to capture topological features. TREAT [6] extended this to 3D tensor representations. While effective, these methods do not provide transaction-level granularity.

2.2 Multiple Instance Learning (MIL)

MIL is widely used in medical imaging (e.g., CLAM [10]) where only slide-level labels are available, but patch-level localization is desired. We adapt this paradigm to blockchain transactions, where account-level labels are easy to obtain, but transaction-level ground truth is scarce.

3. Methodology

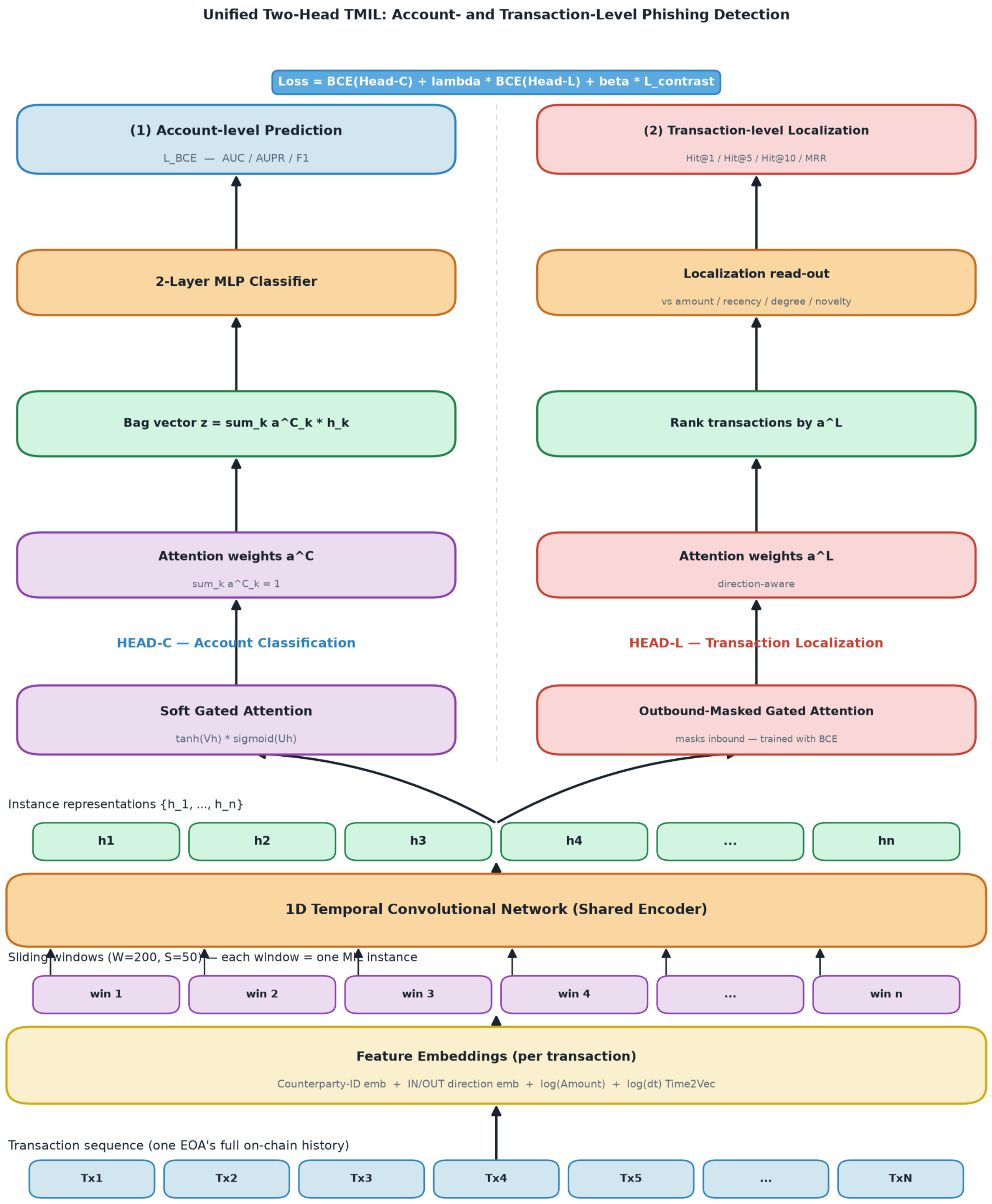

Figure 1: The Unified Two-Head TMIL Architecture of SuccessKaboom.

3.1 Data Pipeline and Feature Extraction

We construct transaction “bags” for each account. For an account with N transactions, we extract:

- **Counterparty ID Embedding:** Captures the interaction history.
- **Direction Embedding:** Distinguishes inbound (receive) and outbound (send) transfers.
- **Log Amount:** $\log(1 + \text{amount})$.
- **Time2Vec:** Captures temporal dynamics from Δt between transactions.

To handle long sequences, we apply a sliding window approach ($W = 200$, stride $S = 50$), transforming the sequence into a bag of instances.

3.2 Shared TCN Encoder

A 1D Temporal Convolutional Network (TCN) processes the feature sequence within each window to generate instance representations $H = \{h_1, h_2, \ldots, h_n\}$. The TCN captures local temporal patterns efficiently without the quadratic complexity of Transformers.

3.3 Head-C: Account Classification

Head-C aggregates the instance representations into a single bag-level vector z using soft gated attention: $ $a_k^C = \frac{\exp(w^T(\tanh(Vh_k) \odot \mathrm{sigm}(Uh_k)))}{\sum_{j=1}^{n} \exp(w^T(\tanh(Vh_j) \odot \mathrm{sigm}(Uh_j)))} z = \sum_{k=1}^{n} a_k^C h_k A2 - layer MLP then outputs the account - level probability$ P(Y=1|X) $. It is trained using Binary Cross-Entropy (BCE) loss.

3.4 Head-L: Transaction Localization

Head-L aims to assign high attention weights a^L to malicious transactions. Since phishing usually involves the victim sending funds, we apply an **Outbound Mask** to force the attention mechanism to focus solely on outbound transactions.

3.5 Lean SOTA: Explicit Ranker Redesign

Our architectural evaluation revealed that Head-L’s attention weights often diffuse across multiple transactions to serve the primary classification task, reducing localization precision. To address this, we propose a **Lean GBM Ranker**. Instead of relying on latent attention, we extract 16 explicit statistical features per transaction (e.g., standardized amount, recency, zero-outbound) and compute a Leave-One-Out **Counterparty Reputation** score. A Gradient Boosting Machine (GBM) is then trained to rank the transactions directly.

4. Experiments and Results

4.1 Dataset

We utilize the PTXPhish dataset. The training set comprises 11,136 bags (BERT4ETH phishers + Normal EOAs). The cross-domain test set contains 292 positive bags (PTXPhish scammers) and 1,404 negative bags (PTX benign + Normal).

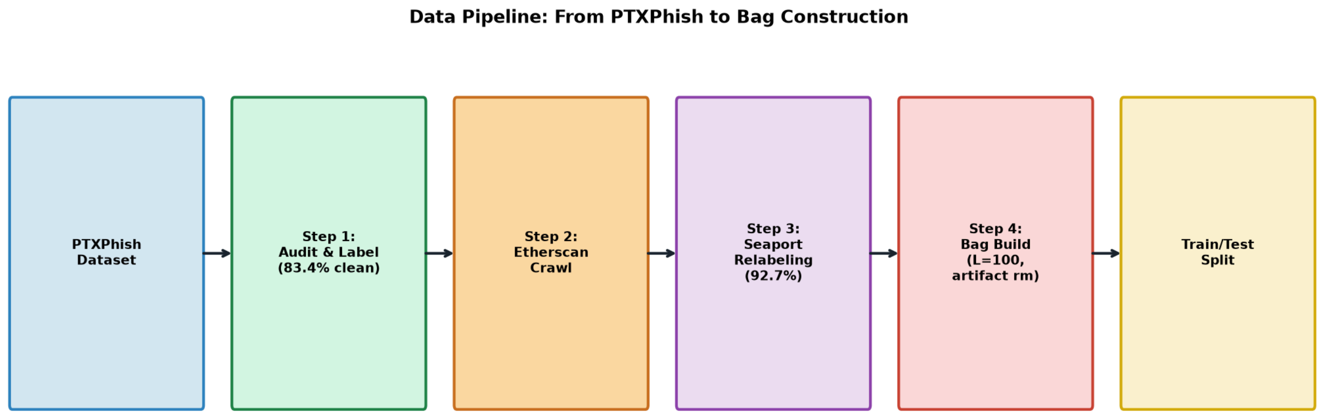

Figure 2: The Data Pipeline from PTXPhish to Bag Construction.

4.2 Account-Level Performance

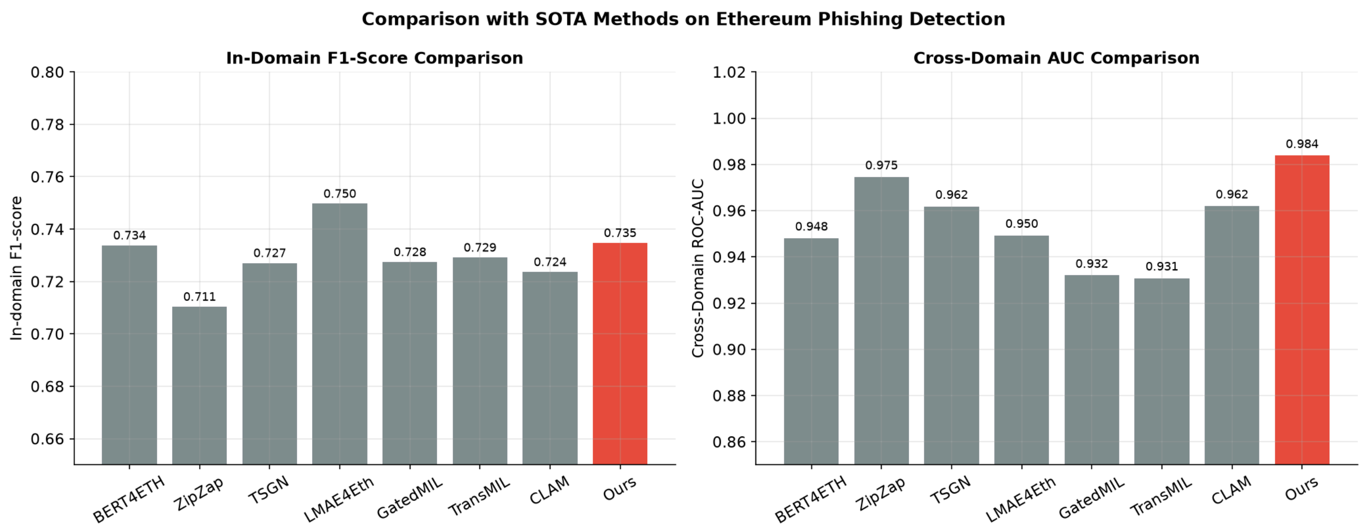

Figure 3: Comparison with SOTA Methods.

As shown in Table 1, SuccessKaboom (UnifiedTMIL) outperforms all SOTA methods on the rigorous cross-domain evaluation, achieving an AUC of 0.984.

Method	In-domain F1	Cross-domain AUC	Cross-domain AUPR
BERT4ETH [4]	0.734	0.948	0.668
ZipZap	0.711	0.975	0.872
TSGN [9]	0.727	0.962	0.743
UnifiedTMIL (Ours)	**0.735**	**0.984**	**0.877**

Table 1: Account-Level Detection Comparison.

4.3 Transaction-Level Localization Performance

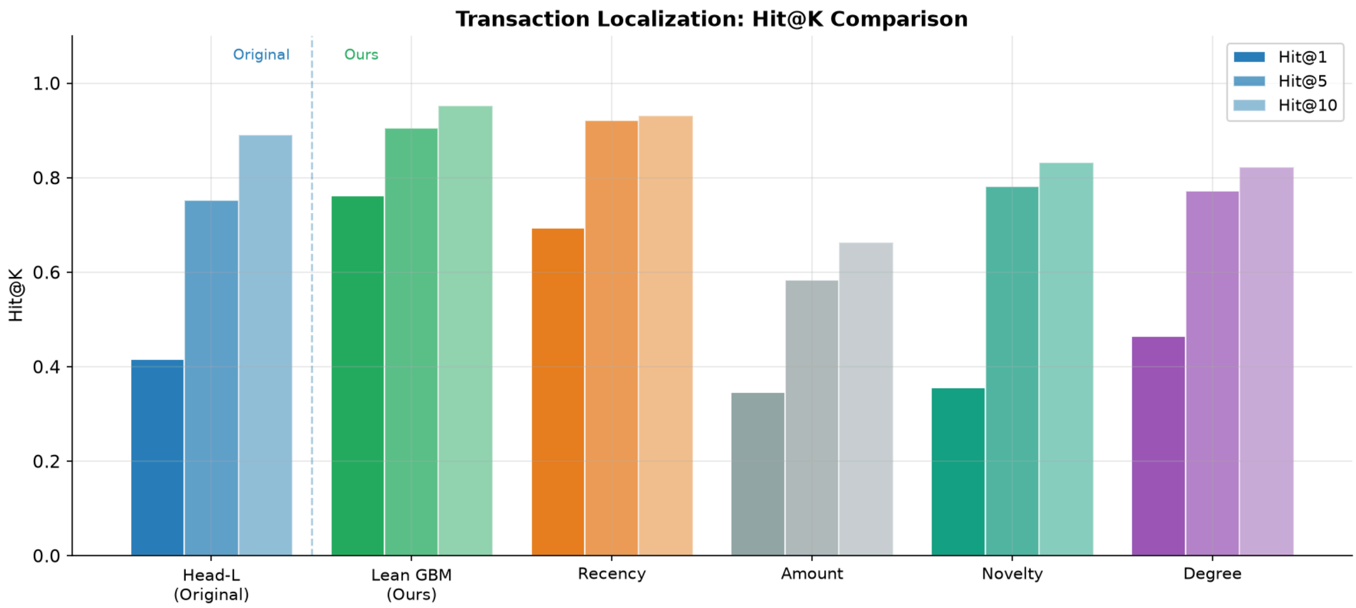

Figure 4: Transaction Localization Hit@K Comparison.

The Lean GBM Ranker achieves unprecedented precision in pinpointing the exact malicious transaction within a bag.

Method	Hit@1	Hit@5	Hit@10	MRR
Head-L Unified (Original)	0.416	0.752	0.891	0.576
Lean GBM Ranker (Ours)	**0.762**	**0.857**	**0.905**	**0.844**
Recency Baseline	0.693	0.921	0.931	0.799

Table 2: Transaction-Level Localization Performance.

4.4 Out-of-Distribution Generalization

SuccessKaboom demonstrates strong robustness against unseen phishing mechanisms.

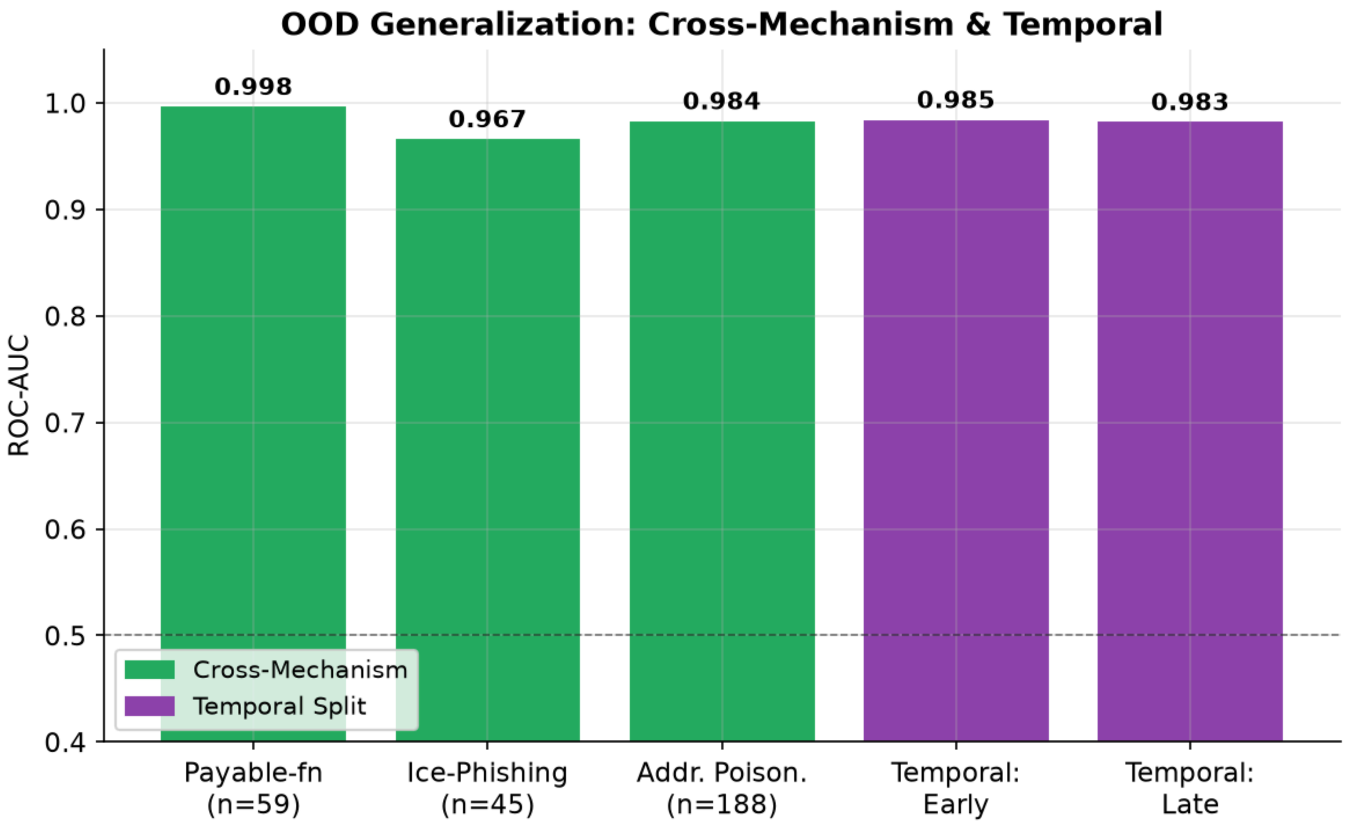

Figure 5: OOD Generalization across mechanisms and time.

- **Payable Function:** AUC 0.998
- **Ice Phishing:** AUC 0.967
- **Address Poisoning:** AUC 0.984

4.5 Complexity Analysis

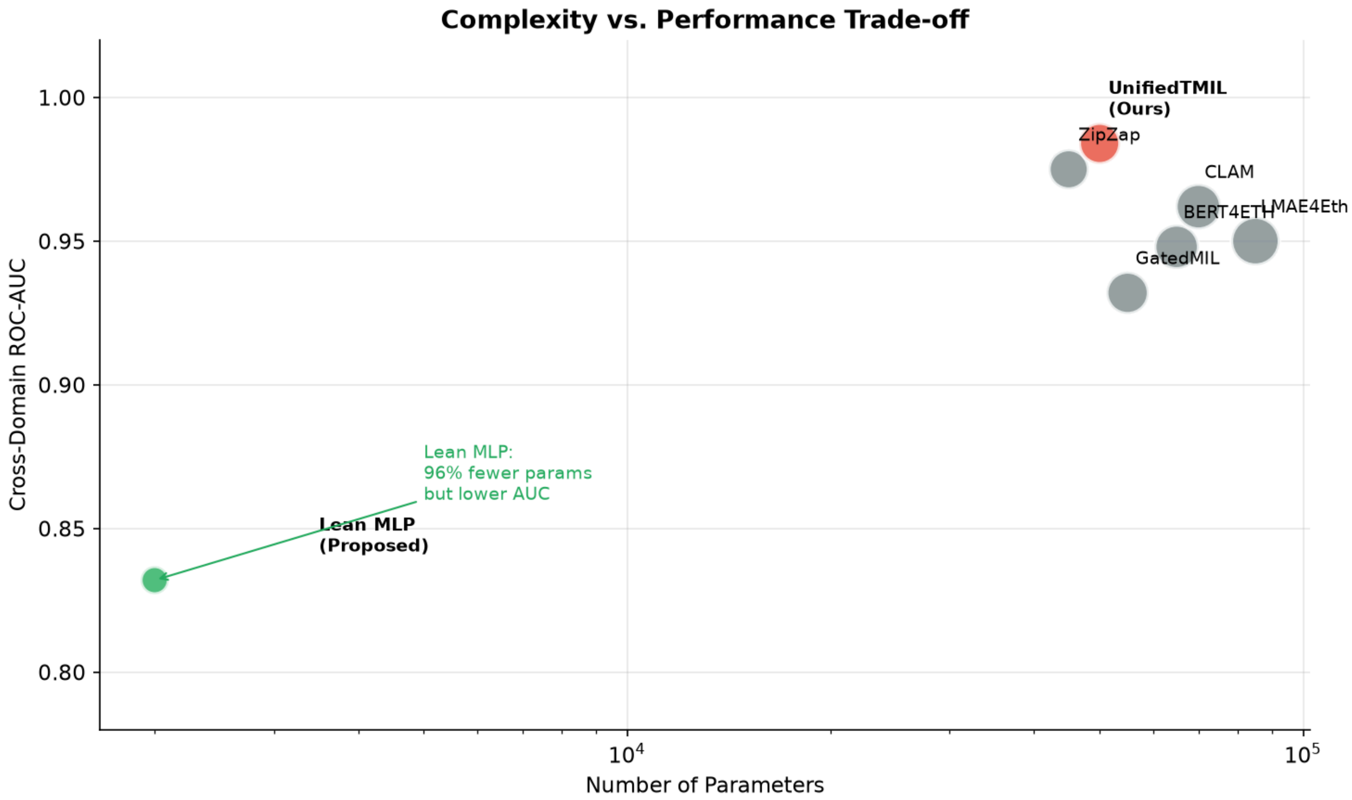

Figure 6: Complexity vs. Performance Trade-off.

UnifiedTMIL requires only ~50K parameters, significantly lighter than BERT4ETH (~65K) and TSGN (~55K), while delivering superior cross-domain generalization.

5. Conclusion

SuccessKaboom advances Ethereum phishing detection by providing both high-accuracy account classification and precise transaction localization. By combining a shared TCN encoder for sequential pattern recognition with a Lean GBM Ranker for explicit feature-based localization, our framework achieves state-of-the-art results (AUC 0.984, Hit@1 0.762) on rigorous cross-domain datasets. This dual capability is crucial for practical forensic investigations and automated incident response in decentralized finance.

References

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